WORKS

Calibrated ZOGY
on DR1 co-adds
69.3% strict injection recovery
91% (99/109) de-
tections explained
JPL Horizons asteroid vetting

HONEST

1.66 null false candidates/field
M42 null & Sgr A* refused
ML dipole biases
openly disclosed
Fail-don't-fake ex-
ternal timeouts

OPEN HORIZONS

Full-mission deep light curves
GPU-accelerated SFFT DIA
Active learning, public votes
IVOA cone-search federation

An Open, Inspectable Software Artefact:

21 429 Runtime Python Lines • 7 611 UI Lines • Netlify Static Edge + Dual Cloud Run + Private GCS
18 Evaluated DR1 Fields • 246 Cutouts Traceable to NASA IRSA • Automated CI Regression Gate

Key Citations: Zackay, Ofek & Gal-Yam 2016 (ZOGY, ApJ 830, 27) • Hu et al. 2022 (SFFT) • Gaia DR3 (A&A 674, A1) • AllWISE / NEOWISE (Mainzer et al.) • SPHEREx Mission (Doré et al.) • NASA/IPAC IRSA SPHEREx Data Release 1 Documentation.